

Large-Area Tactile Sensing via Contact-Induced Modulation of Ultrasonic Waves

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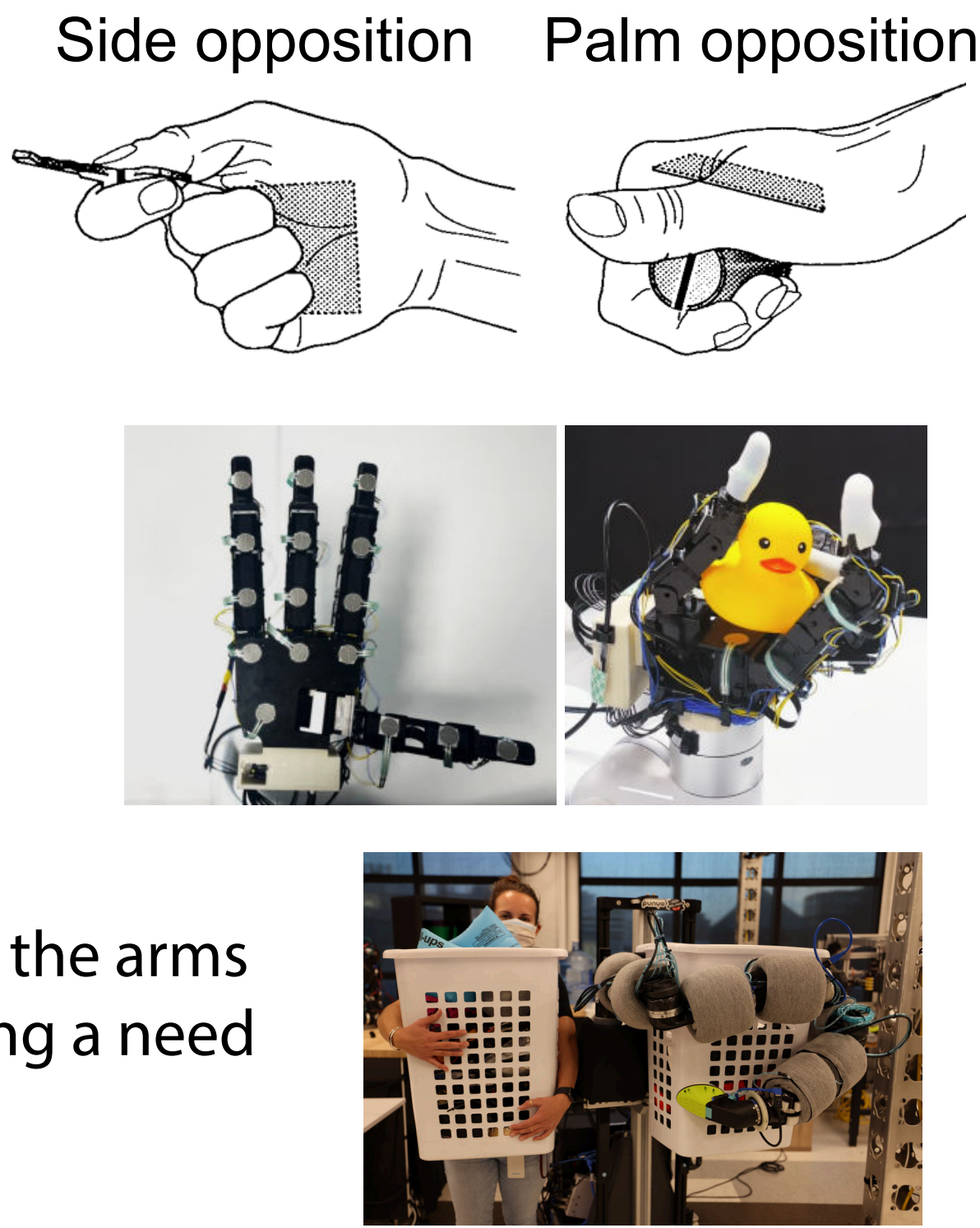
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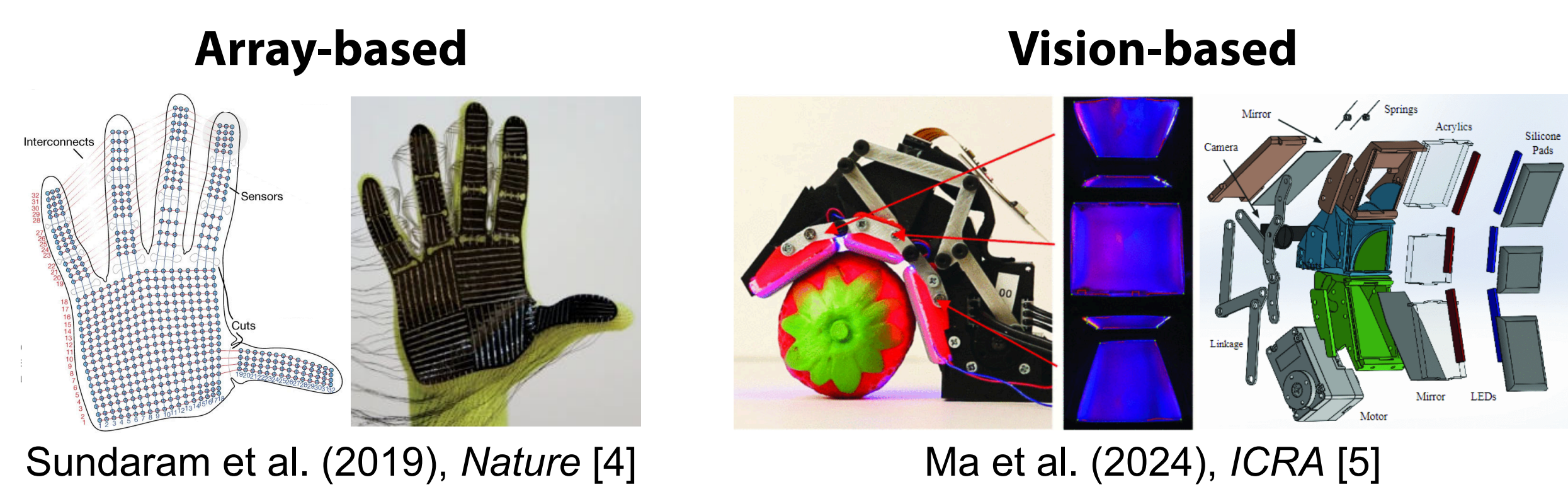
Overview: Tactile sensing is required across increasingly larger portions of robot hands and arms. Here, we introduce an ultrasound-based sensing method that achieves fine localization and accurate normal-force predictions across a large area with few sensing elements.

Distributed tactile sensing facilitates dexterous manipulation

- Many common grasps rely on contact beyond the fingertips, including the palm and sides of the fingers [1].
- Extending tactile sensing beyond the fingertips improves performance in contact-rich tasks [2].
- Broader manipulation tasks can recruit the arms and other body surfaces [3], highlighting a need for scalable sensing architectures.



Existing high-resolution sensing technologies scale poorly with increasing sensor area

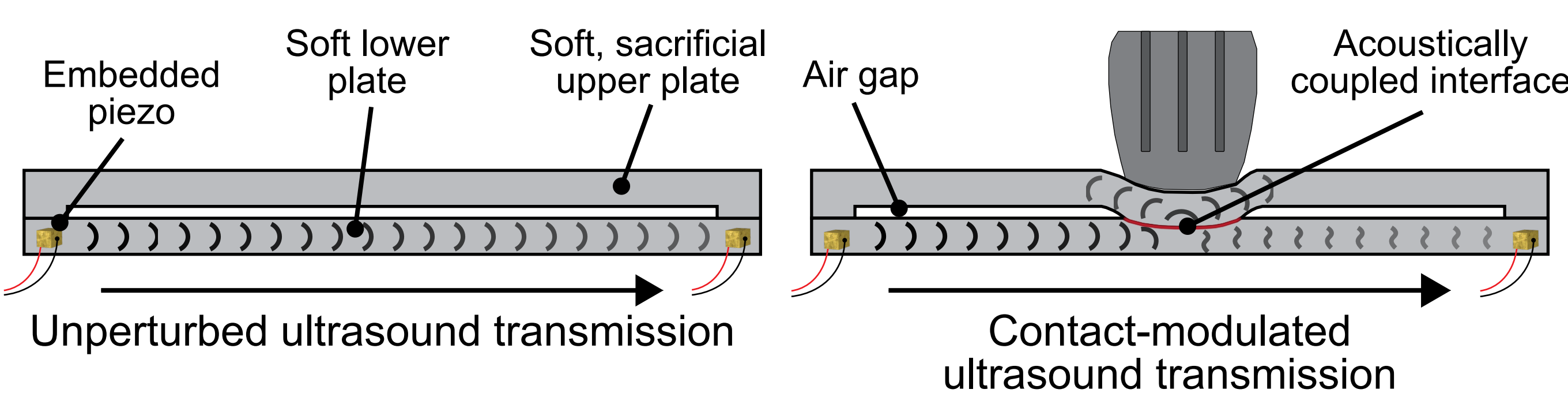


Key challenge: Develop tactile sensing architectures that extend to larger surfaces without dense sensor arrays or bulky imaging hardware.

Active 1 MHz ultrasound is used to remotely investigate a soft, two-plate system

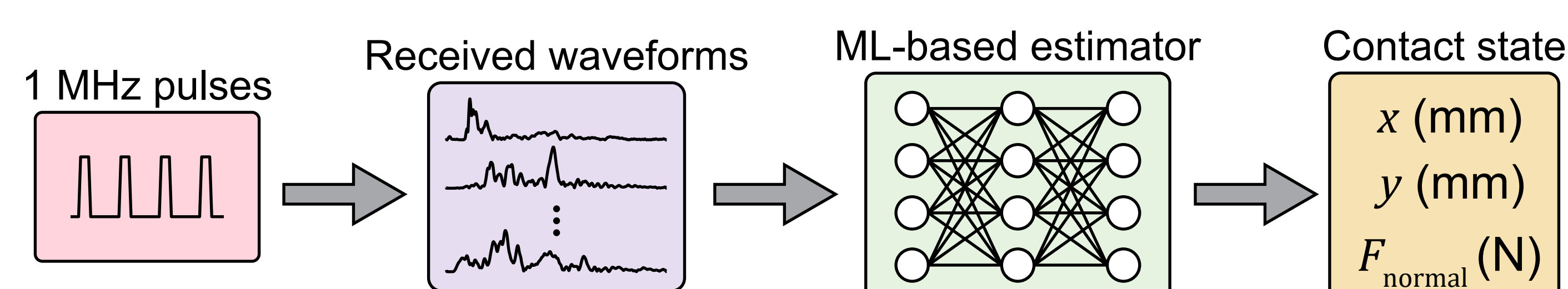
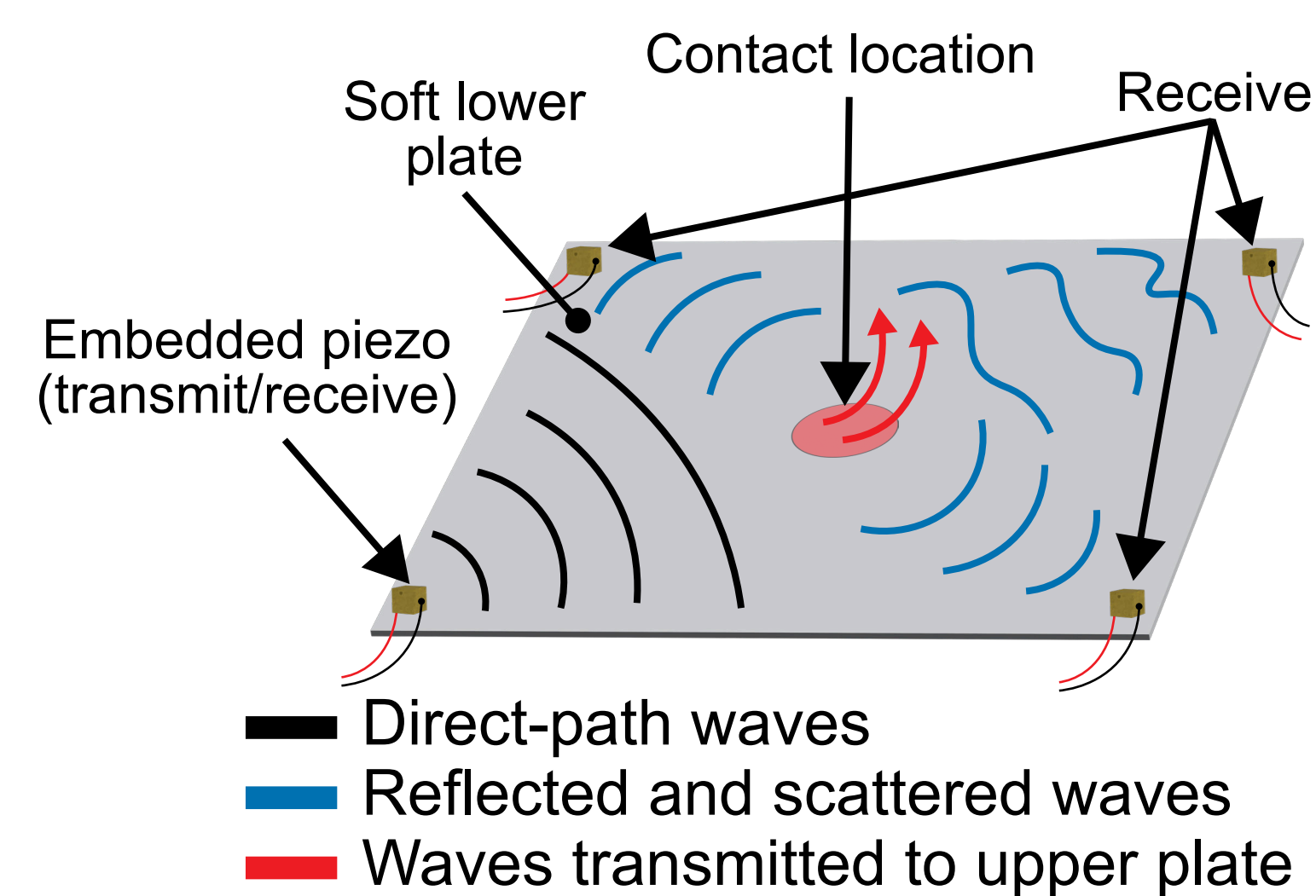
No contact: Plates are acoustically isolated.

External contact: Plates are acoustically coupled.



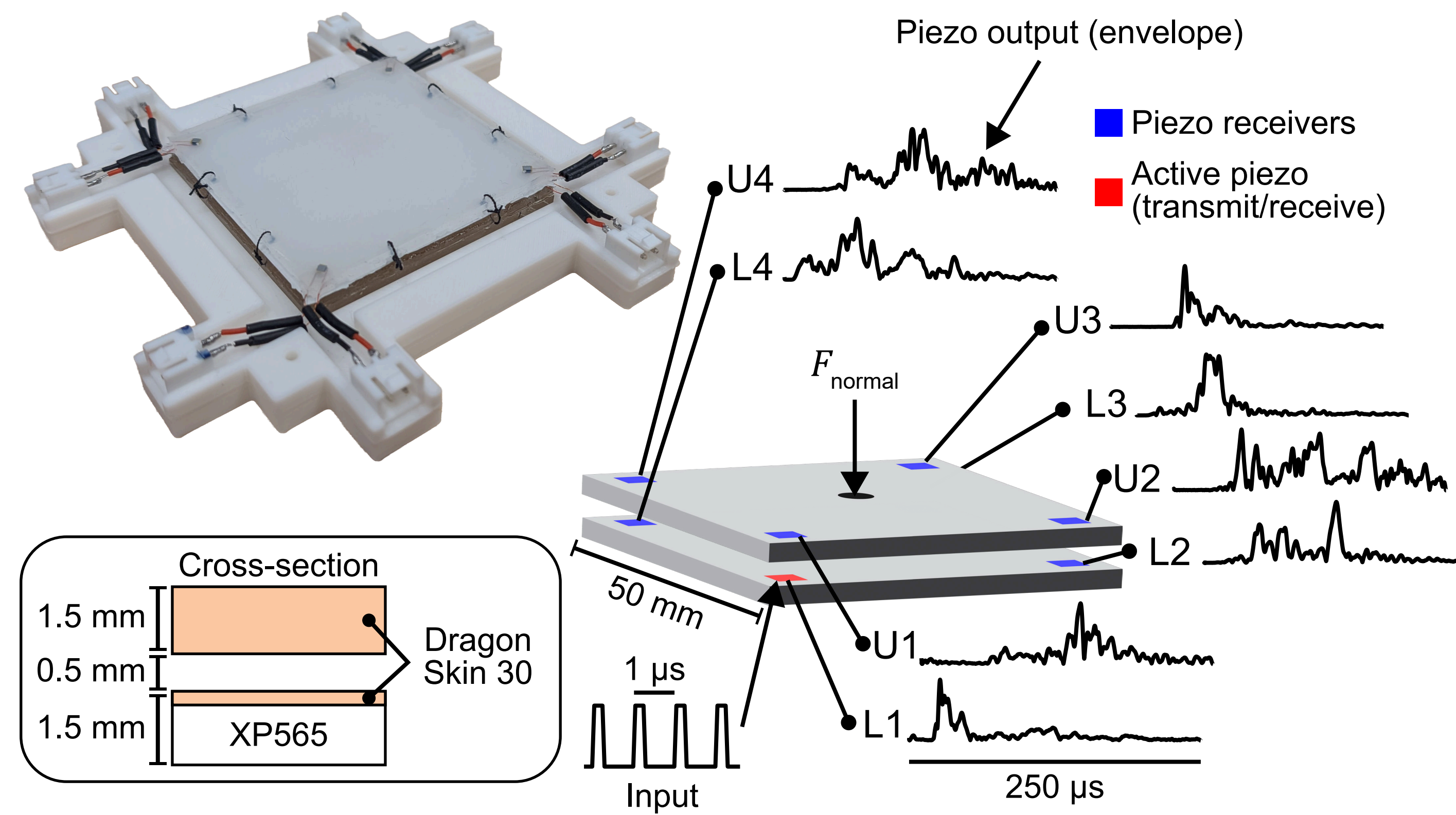
- Contact modifies wave propagation through reflection, scattering, and transmission.

- Materials with low acoustic damping allow waves to interact repeatedly with the contact, producing more distinctive waveforms.

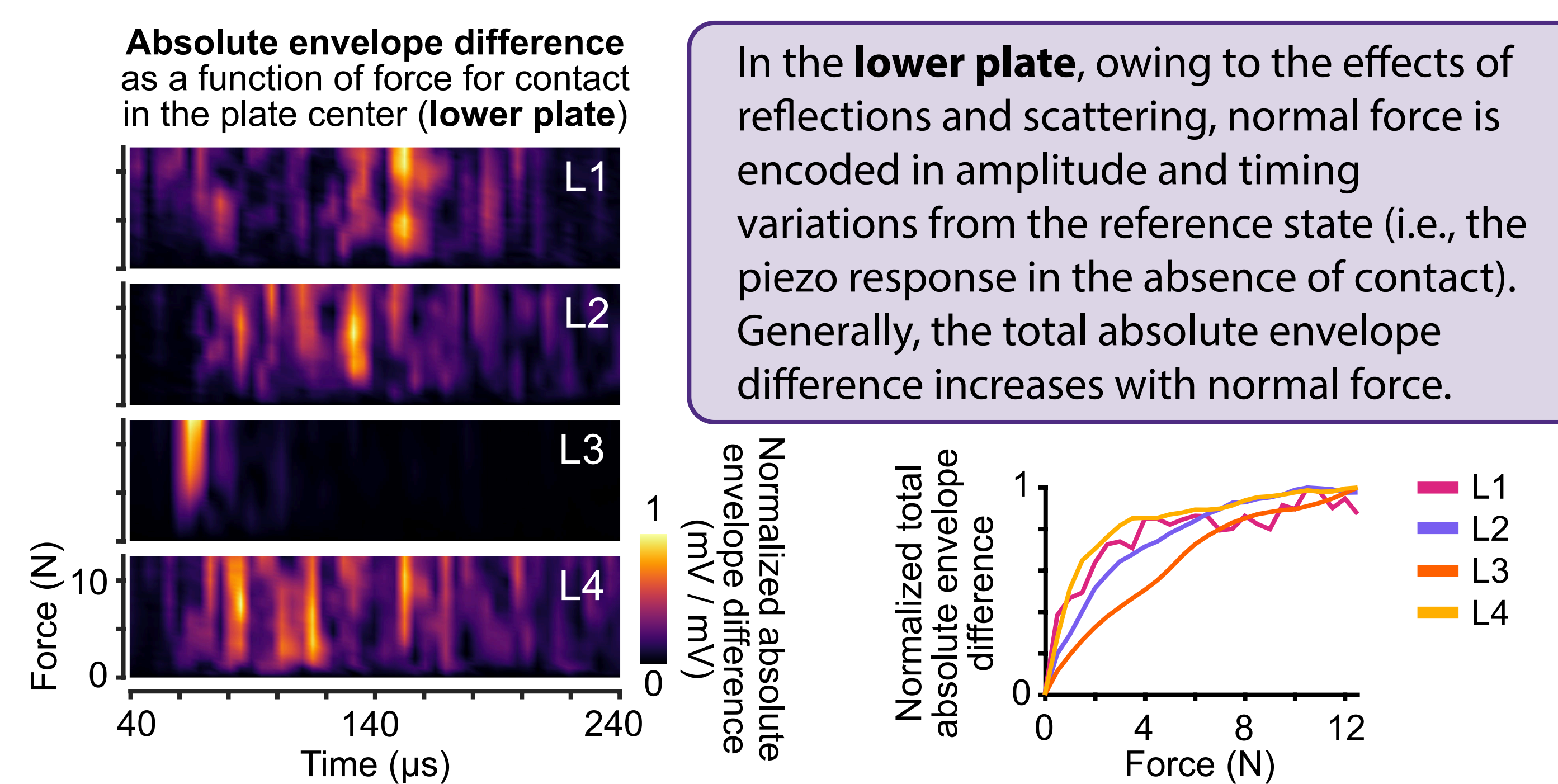
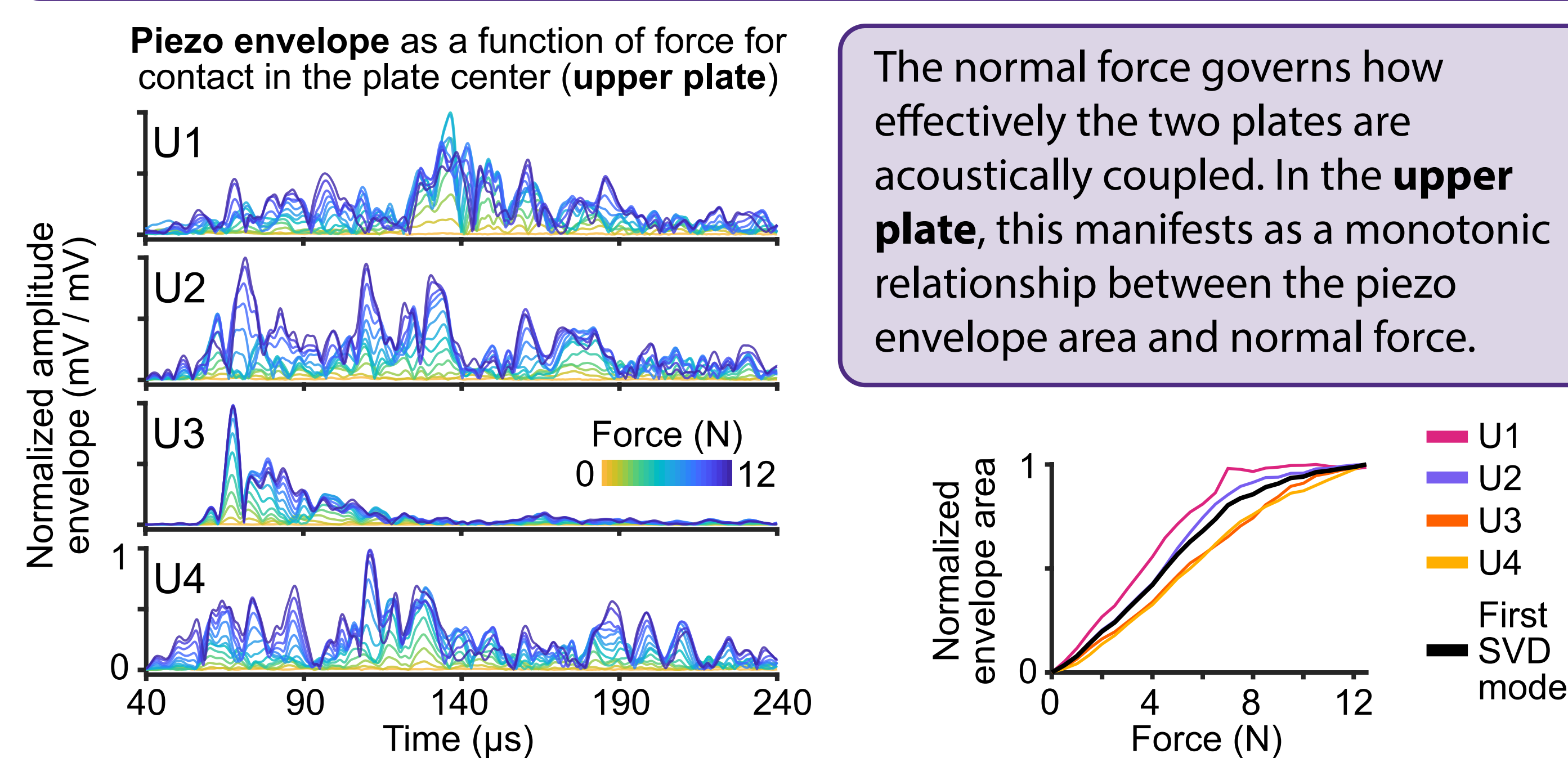


Key idea: Contact-modulated waveforms are used to predict low-dimensional contact states.

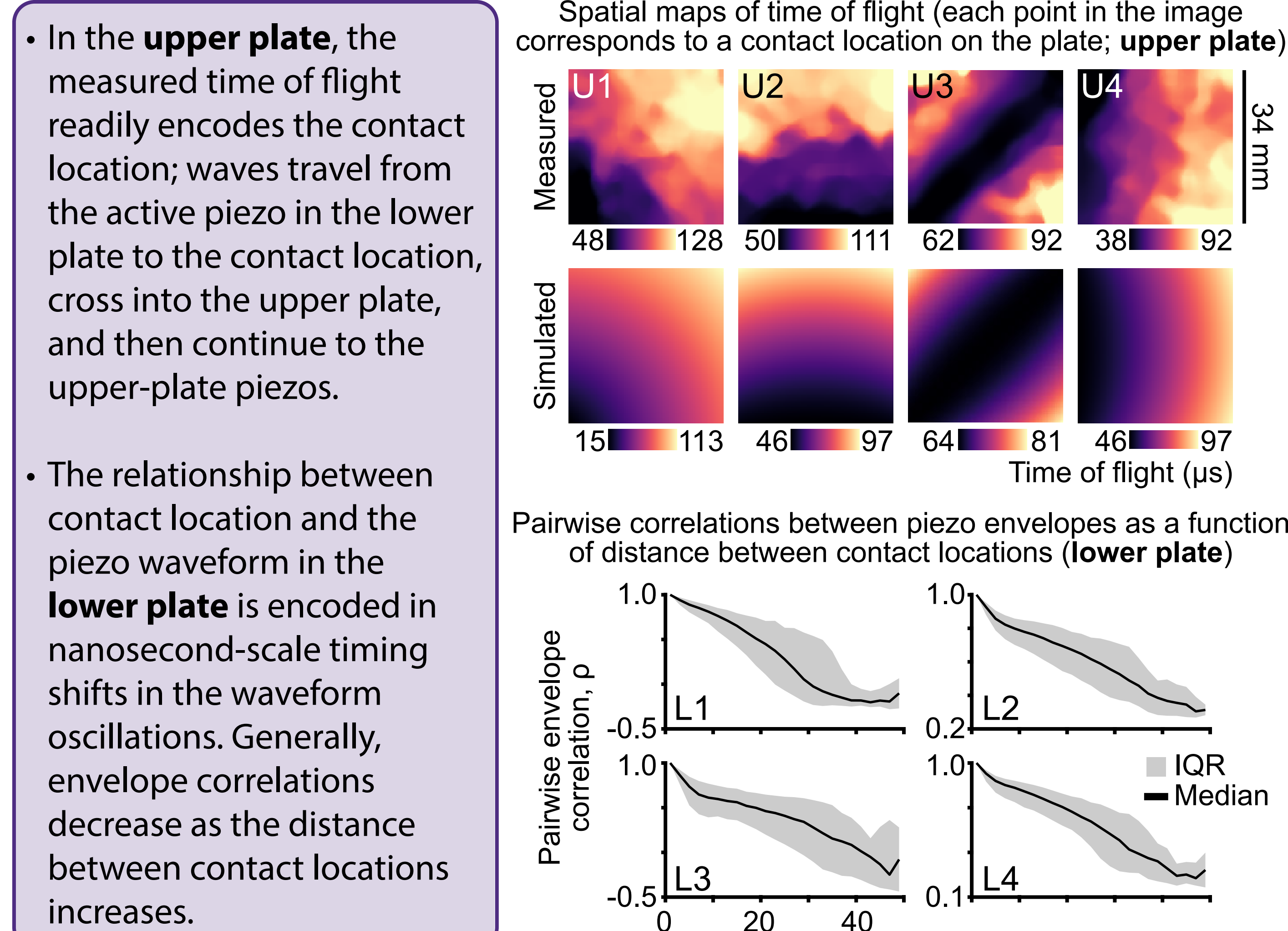
Contact-induced modulations in ultrasound waves encode contact state



How do waveforms encode force?

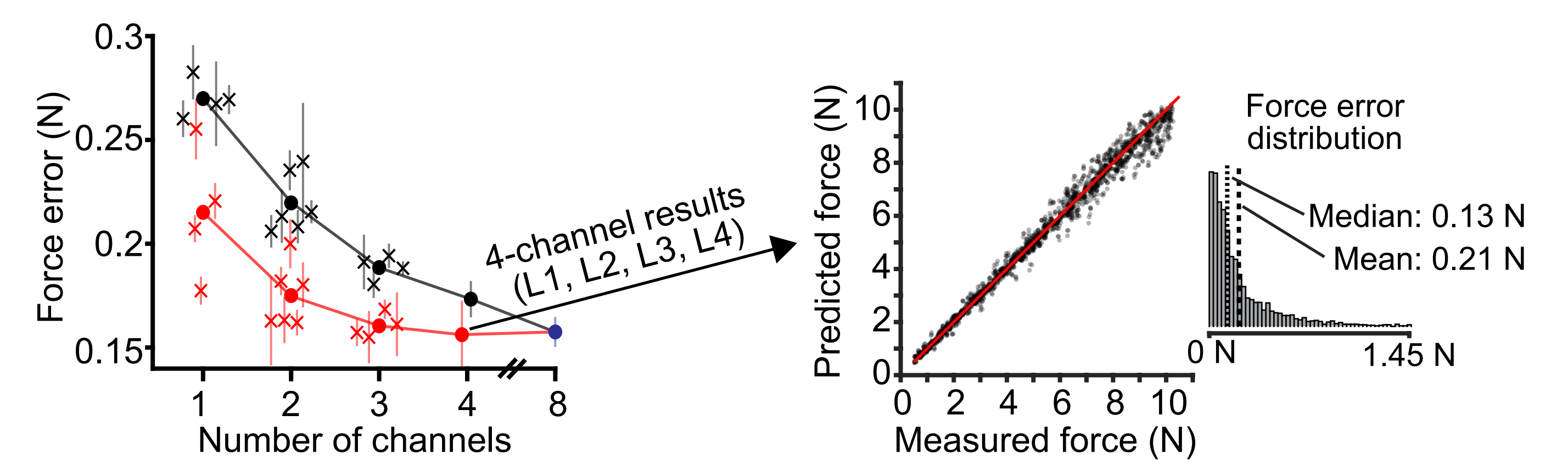
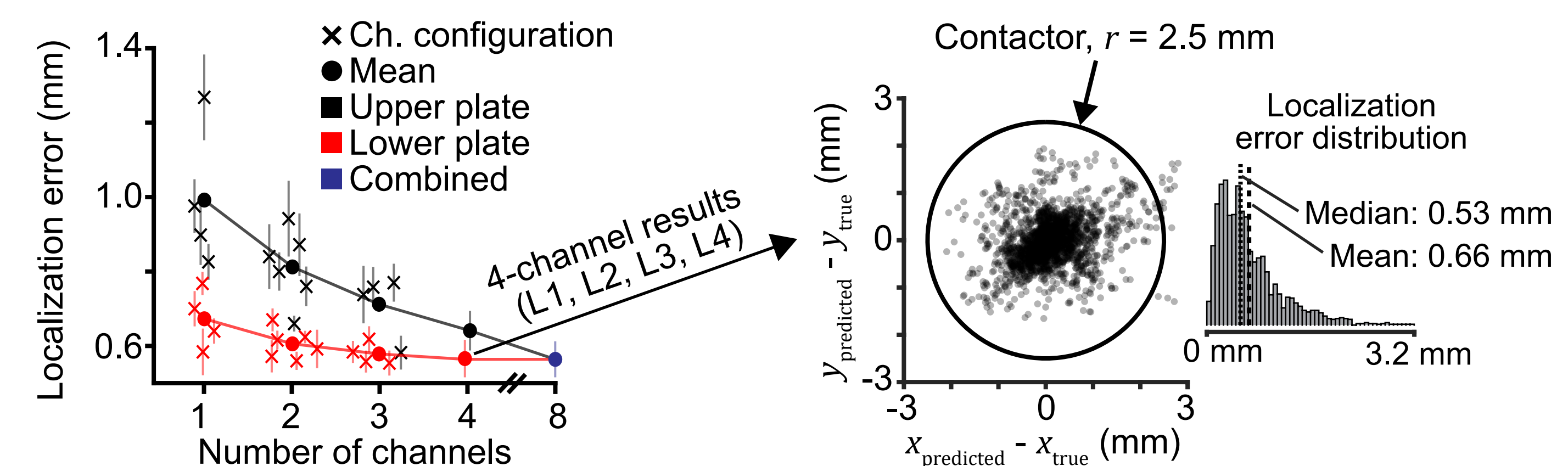
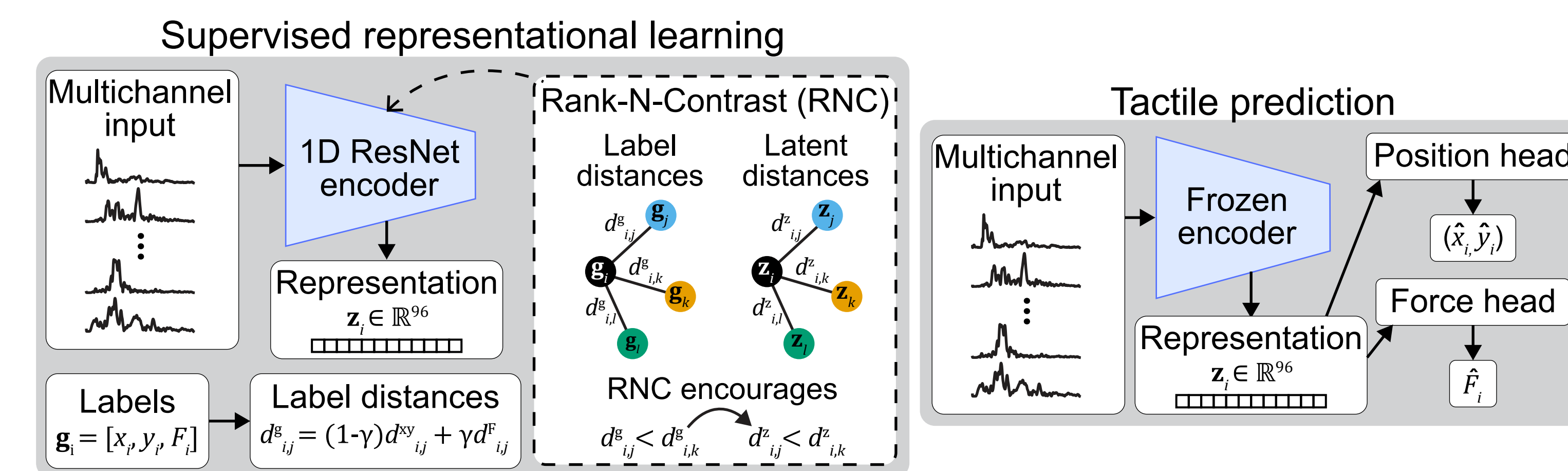


How do waveforms encode contact location?



Contact states can be predicted with high accuracy using supervised learning

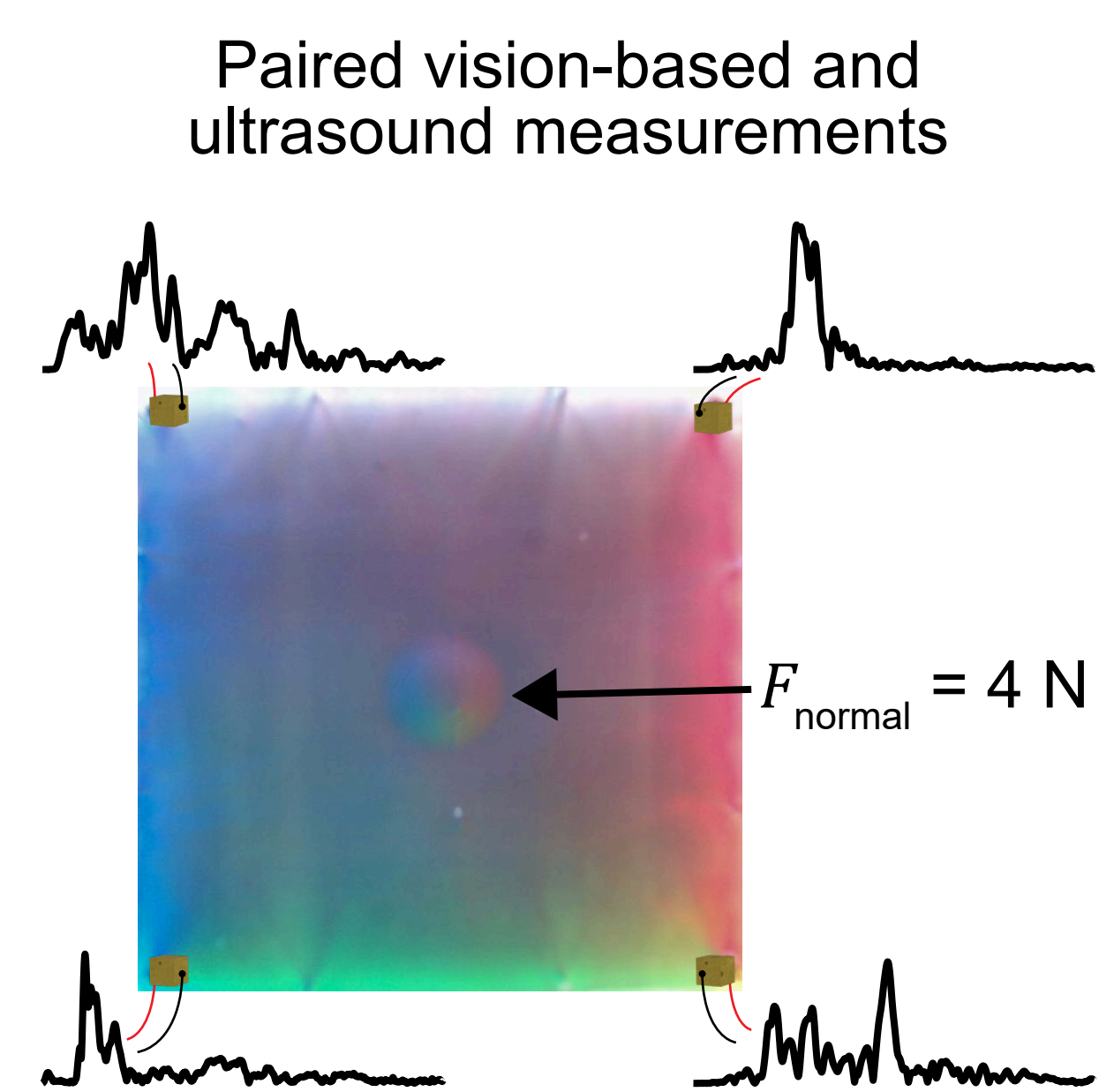
We captured ~8,000 training examples (and ~2,000 unseen test examples) that varied in force and location, and trained a model to predict the contact state.



Key result: We achieved 600 μ m localization and 175 mN normal-force accuracy on average with as few as two sensing elements.

Next steps

- Scalability:** We are currently fabricating a forearm-scale sensor to evaluate how our operating principle scales with increasing sensing area.
- Probe size and shape:** Current training and evaluation used a fixed-radius probe. We plan to extend prediction to probe size and, eventually, shape.
- Cross-modal learning:** We collected vision-based measurements of the plate interface alongside ultrasound measurements. We will explore whether these paired data enable us to predict richer contact states from ultrasound.



References and acknowledgments

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